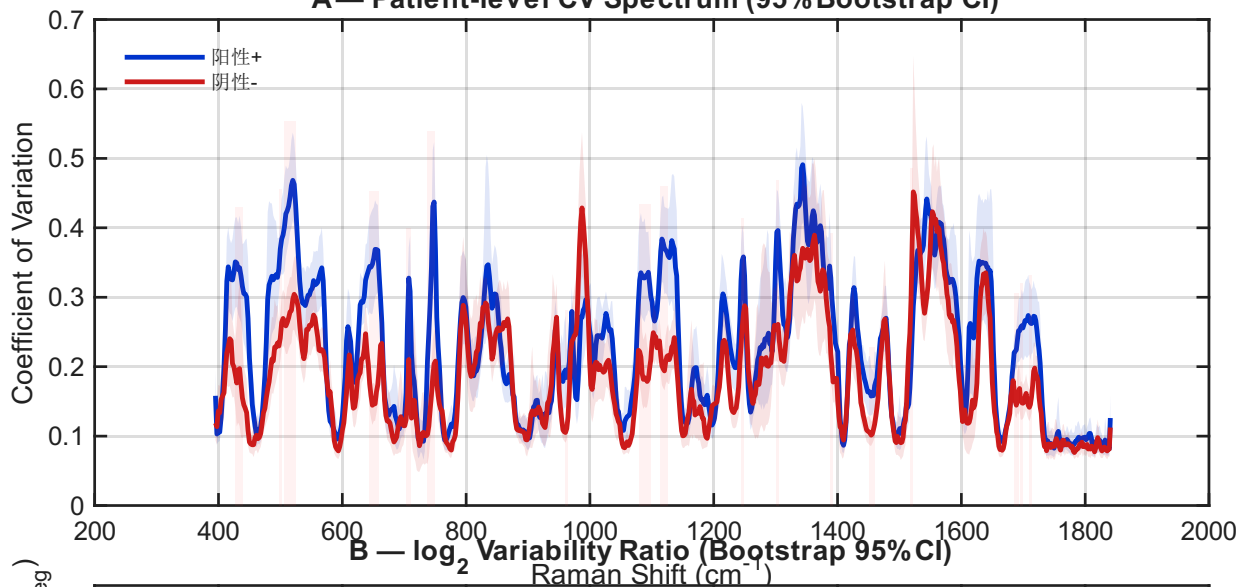
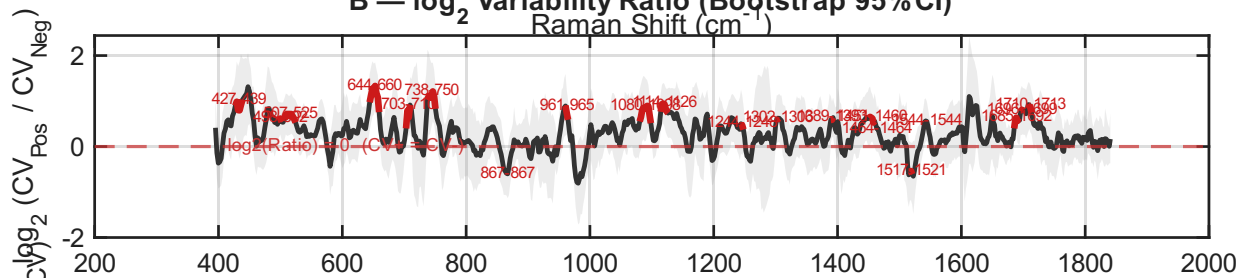


Spectral Variability Analysis v2.1 (Patient-level)

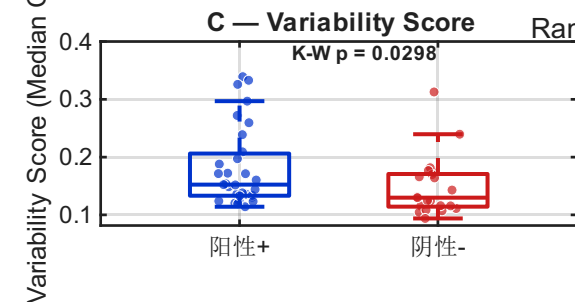
A— Patient-level CV Spectrum (95% Bootstrap CI)



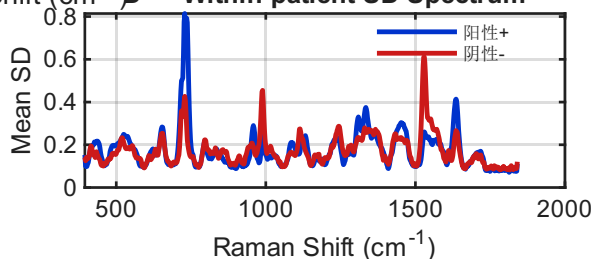
B — $\log_2$ Variability Ratio (Bootstrap 95% CI)



C — Variability Score



Raman Shift (cm⁻¹) **D — Within-patient SD Spectrum**



Band (cm-1)	CV+	CV-	log2R p	Cliff	Assignment	
644 – 660	0.3536	0.1632	+1.12	0.0001	0.65	Band commonly attributed to tyrosine (c-c twisting) vibration
1080 – 1098	0.3263	0.1943	+0.75	0.0001	0.64	Band commonly attributed to po ₂ symmetric stretch (dna/rna, phospholipid) vibration
1114 – 1126	0.3702	0.2065	+0.84	0.0002	0.61	Band commonly attributed to c-n stretching (protein backbone) vibration
738 – 750	0.3326	0.1588	+1.07	0.0003	0.60	Band commonly attributed to tryptophan (indole ring) vibration
498 – 502	0.3648	0.2448	+0.58	0.0003	0.59	Unknown
507 – 525	0.4386	0.2799	+0.65	0.0004	0.59	Unknown
1451 – 1460	0.1624	0.1070	+0.60	0.0004	0.59	Band commonly attributed to ch ₂ scissoring (protein, lipid) vibration
427 – 439	0.3382	0.1813	+0.90	0.0004	0.59	Unknown
1389 – 1393	0.3009	0.1982	+0.60	0.0005	0.58	Band commonly attributed to coo ⁻ symmetric stretch (eps, biofilm matrix) vibration
703 – 710	0.2647	0.1620	+0.71	0.0005	0.57	Unknown
1685 – 1692	0.2314	0.1573	+0.56	0.0007	0.56	Unknown
867 – 867	0.1774	0.2689	-0.60	0.0008	0.55	Band commonly attributed to proline / tyrosine (c-c stretch) vibration
961 – 965	0.1993	0.1793	+0.59	0.0009	0.54	Band commonly attributed to c-h stretching (lipid unsaturation) vibration

Effect Size Spectrum (Bootstrap 95% CI on sig. regions)

